

Brock Biology of Microorganisms, 15e (Madigan et al.)
Chapter 20 Microbial Ecosystems

20.1 Multiple Choice Questions

1) Which metric describes the proportion of EACH species present in a community?

- A) species abundance
- B) species diversity
- C) species evenness
- D) species richness

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.1

2) Which of the following represents increasing microbial diversity?

- A) guild-community-population
- B) population-guild-community
- C) cell-community-population
- D) community-guild-population

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 20.2

3) The abundance of cyanobacterial mats has greatly declined due to the evolution of

- A) antimicrobial-producing fungi.
- B) bacteriophages.
- C) animal grazers.
- D) *Proteobacteria*.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.5

4) A notorious biofilm former implicated in cystic fibrosis is

- A) *Bacillus subtilis*.
- B) *Escherichia coli*.
- C) *Lactobacillus acidophilus*.
- D) *Pseudomonas aeruginosa*.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.4

5) Some nitrifying prokaryotes carry out metabolic cooperation by a process known as

- A) competition.
- B) niche partitioning.
- C) syntrophy.
- D) antagonism.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.3

6) The generation time of *Escherichia coli* in the human intestine is _____ in laboratory culture.

- A) at about the same rate as
- B) faster than
- C) more optimal than
- D) slower than

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.3

7) In a stratified lake, the warmer and less dense layer is called the

- A) epilimnion.
- B) hypolimnion.
- C) hyperlimnion.
- D) thermocline.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.8

8) The photosynthetic pigments bacteriorhodopsin and proteorhodopsin are present in _____ and _____, respectively.

- A) *Archaea* / *Bacteria*
- B) *Bacteria* / *Archaea*
- C) *Bacteria* / *Bacteria*
- D) *Bacteria* / *Eukarya*

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.11

9) In the soil, oxygen concentrations are highly variable even within a soil particle. The center of the particle may be anoxic even if the outer portions are oxygen rich. In this context, facultative anaerobes could be found

- A) at the surface of the particle.
- B) in the center of the particle.
- C) in the middle regions of the particle.
- D) throughout the particle.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.3

10) A population consists of

- A) all of the individuals of all species in the same area.
- B) all of the individuals of one species in the same area.
- C) all of the individuals that interact in different areas.
- D) all of the individuals that interact in the same area.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.1

11) Biofilm structure is determined by

- A) nutritional factors.
- B) signaling factors.
- C) the local environment.
- D) the local environment, nutritional factors, and signaling molecules.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.4

12) Which metabolic strategy is most common in chemolithotrophic mats?

- A) ammonia oxidation
- B) iron oxidation
- C) nitrate reduction
- D) sulfur oxidation

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.5

13) During periods of stratification, transfer between surface and bottom waters is controlled by

- A) mixing.
- B) diffusion.
- C) niche partitioning.
- D) current.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.8

14) Assuming exposure to the same climate, which of the following aquatic systems would be most stratified?

- A) 20 m deep freshwater river
- B) 50 m deep marine water fjord
- C) 200 m deep saltwater lake
- D) 300 m deep freshwater lake

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 20.8

15) Which is the correct order of increasing size in soil particles?

- A) clay > sand > silt
- B) clay > silt > sand
- C) sand > clay > silt
- D) silt > clay > sand

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 20.6

16) A soil that lacks a dominate particle size is referred to as a(n)

- A) aggregation.
- B) bisequum.
- C) loam.
- D) mottled soil.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.6

17) According to bacterial abundance studies done so far on soils, the _____ represent the most abundant phylum.

- A) *Proteobacteria*
- B) *Acidobacteria*
- C) *Actinobacteria*
- D) *Firmicutes*

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.6

18) Which bacterial genus is responsible for producing ferrous iron (Fe^{2+}) in groundwater?

- A) *Alteromonas*
- B) *Aquifex*
- C) *Geobacter*
- D) *Loktanella*

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.7

19) An organism living in the bottommost region of a body of water is described as being

- A) benthic.
- B) litoral.
- C) neritic.
- D) pelagic.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.8

20) The chemical oxygen demand (COD) of a body of water is determined using a(n)

- A) microelectrode.
- B) oxygenator.
- C) spectrophotometer to determine the OD_{540nm}.
- D) strong oxidizing agent.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.8

21) Which aquatic habitat contains a pelagic zone?

- A) estuary
- B) lake
- C) salt marsh
- D) open ocean

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.9

22) Which are the most dominant chemolithotrophs in pelagic waters?

- A) ammonia-oxidizing *Archaea*
- B) ammonia-oxidizing *Bacteria*
- C) sulfate-reducing *Bacteria*
- D) *Prochlorococcus*

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.9

23) From the aquatic systems below, where are heterotrophic *Bacteria* the most abundant?

- A) marine coastal waters
- B) deep sea
- C) Sargasso Sea
- D) open ocean

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 20.9

24) All *Prochlorococcus* strains contain the pigments

- A) chlorophylls *a* and *b*, but not phycobilins.
- B) chlorophyll *a* and phycobilins, but not chlorophyll *b*.
- C) chlorophyll *b* and phycobilins, but not chlorophyll *a*.
- D) chlorophylls *a* and *b* and phycobilins.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.10

25) Which of the following is NOT an oxygenic phototrophic microorganism present in marine waters?

- A) *Ostreococcus*
- B) *Prochlorococcus*
- C) *Roseobacter*
- D) *Trichodesmium*

Answer: C

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 20.10

26) How does the concentration of prokaryotic cells at the surface of pelagic water compare with the concentrations at depths below 1000 m?

- A) There is a decrease in concentration from 1 m to 1000 m in depth, then a dramatic increase below 1000 m.
- B) There is a decrease in concentration with increasing depth.
- C) There is a steady increase in concentration with increasing depth.
- D) There is an increase in concentration at depths below 1000 m.

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 20.9

27) *Epsilonproteobacteria* are most dominant in hydrothermal vents because they

- A) reduce nitrogen.
- B) oxidize sulfide and sulfur as electron donors.
- C) use oxygen as electron acceptors.
- D) bind divalent cations.

Answer: B

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 20.14

28) In deep pelagic waters, the archaeal species present are almost all members of the

- A) *Crenarchaeota*.
- B) *Euryarchaeota*.
- C) *Nanoarchaeota*.
- D) *Thaumarchaeota*.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.11

29) Phylogenetic analyses of marine prokaryotic diversity in both shallow and deep-sea marine sediments have found that both types of sediments are dominated by

- A) *Firmicutes*.
- B) *Planctomycetes*.
- C) *Proteobacteria*.
- D) *Chloroflexi*.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.13

30) Which statement is TRUE about the fluid from the hydrothermal vents?

- A) The fluid contains large amounts of oxidized inorganic materials.
- B) The fluid contains large amounts of reduced inorganic materials.
- C) The fluid contains large amounts of organic material.
- D) The fluid usually contains about equal amounts of organic and inorganic material.

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 20.14

31) The primary metal sulfides emitted from the black smokers are _____ sulfides.

- A) copper
- B) iron
- C) magnesium
- D) zinc

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.14

32) What adaptation allows organisms to tolerate high pressures?

- A) decreased cell size
- B) decreased lipid content
- C) increased ribosome size
- D) increased unsaturated fatty acids

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.12

33) A chemical that comes from outside the ecosystem is referred to as a(n) _____ chemical.

- A) synthropic
- B) lithotrophic
- C) allochthonous
- D) pleomorphic

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.2

34) Ecological theory states that for every organism there is at LEAST one _____, and the microenvironment where the organism is most successful is called the _____.

- A) population / community
- B) niche / prime niche
- C) prime niche / niche
- D) community / population

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.3

35) A population of metabolically related microorganisms is called a(n)

- A) guild.
- B) niche.
- C) ecosystem.
- D) haplotype.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.2

36) Which of the following is an example of a biofilm?

- A) transient skin microbes
- B) dental plaque
- C) marine diatoms
- D) a planktonic microcolony

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 20.4

37) Sedimentation in bogs and marshes develop _____ soils.

- A) inorganic
- B) mineral
- C) organic
- D) loamy

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.6

38) The number of *Roseobacter* in the ocean describes

- A) species diversity.
- B) species richness.
- C) species abundance.
- D) species prevalence.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.1

39) _____ are the main primary producers in freshwater environments.

- A) Phototrophic microorganisms
- B) Heterotrophic microorganisms
- C) Chemolithotrophic microorganisms
- D) Viral microorganisms

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.8

40) Sudden nutrient perturbations in an environment generally _____ the overall prokaryotic diversity.

- A) increase
- B) decrease
- C) maintain
- D) replicate

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.6

41) Water in the rocks and soils of Earth's subsurface is called

- A) groundwater.
- B) surface water.
- C) subsurface water.
- D) deep water.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.7

42) The microbial oxygen-consuming capacity of a water sample is called its

- A) chemical oxygen demand.
- B) biochemical capacity.
- C) biochemical oxygen demand.
- D) chemical capacity.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.8

43) Two major environmental extremes of the deep sea are high _____ and low _____.

- A) pressure / nutrient levels
- B) nutrient levels / pressure
- C) temperature / pressure
- D) oxygen / pressure

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.12

44) Biomass generation due to heterotrophic organisms is called _____ production.

- A) primary
- B) secondary
- C) auxotrophic
- D) regulatory

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.11

45) In the oceans, prokaryotes are greatly outnumbered by

- A) zooplankton.
- B) diatoms.
- C) viruses.
- D) phytoplankton.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.11

46) _____ use inorganic electron donors released from deep-sea hydrothermal vents.

- A) Chemolithotrophs
- B) Photoheterotrophs
- C) Chemoheterotrophs
- D) Photolithotrophs

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 20.14

47) Cells within a biofilm excrete an adhesive matrix primarily composed of _____ but can also contain nucleic acids as well as proteins.

- A) lipids
- B) fatty acids
- C) monomers
- D) polysaccharides

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.4

48) Cyclic dimeric guanosine monophosphate (c-di-GMP) serves as a messenger molecule in bacterial cells to coordinate a shift from _____ to _____ growth.

- A) sessile / planktonic
- B) planktonic / sessile
- C) monomorphic / polymorphic
- D) stationary / exponential

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 20.4

49) The most abundant prokaryotic organism in the ocean is of the _____ genus.

- A) *Pelagibacter*
- B) *Puniceispirillum*
- C) *Roseobacter*
- D) *Prochlorococcus*

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.11

20.2 True/False Questions

1) An ecosystem includes both the living organisms and abiotic materials that function together.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.1

2) Many microbial habitats are unsuitable for plants and animals.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.3

3) The physiochemical conditions of a microenvironment, by definition, remain static.

Answer: FALSE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 20.3

4) A microbial population can contain several microbial communities.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.1

5) Chemoheterotrophic bacteria are primary producers.

Answer: FALSE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 20.5

6) Cyclic di-guanosine monophosphate is upregulated as *Pseudomonas aeruginosa* cells are declining.

Answer: FALSE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 20.4

7) Arid soils are among the most extreme environments on earth.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.6

8) Due to a vast array of nutrients, microorganisms are often capable of sustaining exponential growth in the environment.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.3

9) The cells within a biofilm can undergo intra-species signaling; however, inter-species signaling does not occur.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.4

10) Mineral soils largely come from weathered limestone and sand.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.6

11) The formation of soil involves a combination of biological, chemical, and physical processes.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.6

12) Soil particles are classified based on their size.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.6

13) Chemoorganotrophic bacteria can facilitate the formation of soil through the secretion of organic acids.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.6

14) Limiting nutrients in soils are generally organic compounds.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.6

15) Functional differences can be observed by using 16S rRNA gene phylogenetics of a microbial community.

Answer: FALSE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 20.6

16) Microbial life has been identified at depths of 3 km despite being anoxic.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.7

17) All stratified lakes contain a thermocline.

Answer: TRUE

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 20.8

18) *Gammaproteobacteria* can be used for biodegradation of hydrocarbons.

Answer: TRUE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 20.9

19) In many marine systems, the carbon and oxygen cycles are directly proportional to each other.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.8

20) Low return of nutrients in the pelagic zone results in relatively low primary production in the open ocean when compared to freshwater lake systems.

Answer: TRUE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 20.9

21) Unlike other cyanobacteria, *Prochlorococcus* lacks phycobilins.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.10

22) Purple phototrophic bacteria that use bacteriochlorophyll *a* are oxygenic.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.10

23) *Archaea* and *Bacteria* dominate the upper 100 m of the open ocean while viruses dominate depths greater than 100 m.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.11

24) Oligotrophic organisms do NOT grow well under nutrient-replete conditions.

Answer: TRUE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 20.9

25) In the context of nutrients, a sink is one that slowly releases a particular nutrient over time.

Answer: FALSE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 20.6, 20.9

20.3 Essay Questions

1) Explain three reasons why bacteria form biofilms.

Answer: Biofilms allow bacteria to live in close association with each other, which can be beneficial for nutrient and genetic exchange. They are also a means of microbial self-defense that increase survival. Another advantage of cells being in a biofilm is that they remain in a niche through attachment.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.4

2) Distinguish an ecotype from a species, and provide an example of an ecotype.

Answer: An ecotype is a genetic variant of a species that shares a physiological trait unlike other ecotypes. Ecotypes are distinguished from one another by occupying different niches. For example, one *Prochlorococcus* ecotype is exclusively found in low light conditions, while another *Prochlorococcus* ecotype is present only in high light conditions.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 20.10

3) What is the difference between an organism being piezophilic and piezotolerant?

Answer: An organism that is philic grows best under a certain condition, so piezophilic organisms grow best at a pressure greater than 1 atmosphere. Tolerant organisms are capable of withstanding a particular condition but do not necessarily grow best under that condition. Piezotolerant organisms can tolerate pressures beyond 1 atmosphere, but they grow best at 1 atmosphere.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 20.12

4) Describe three molecular effects high pressure has on the physiology of organisms.

Answer: All of the following are core metabolic processes that are affected by high pressure: DNA synthesis, nutrient transport, protein (enzyme) interactions, and protein synthesis.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.12

5) Provide a reason for the ecological importance of pelagic *Bacteria* and *Archaea*.

Answer: They consume dissolved organic carbon in the oceans, one of the largest pools of usable organic carbon on Earth.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 20.11

6) Explain how both biochemical oxygen demand and chemical oxygen demand are measured.

Answer: The biochemical oxygen demand is determined by highly oxygenating a water sample. The water is sealed in a bottle and incubated for 5 days at 20°C in the dark. During this time, the organisms should not be limited by oxygen, so a respiration capacity is determined based on the quantity of organic material in the water sample. Chemical oxygen demand is measured by oxidizing all organic material into CO₂ by using an oxidizing agent, and the loss of oxidizing agent is quantified to infer CO₂ and therefore organic material concentration.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.8

7) Describe two roles viruses have in the open ocean.

Answer: Due to the nature of lysogeny, viruses have a major role in transferring genetic material as viral shuttles. During their lytic cycle, viruses also free up large quantities of nutrients from the dead cells.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.11

8) What advantage do proteorhodopsin-containing organisms have over other heterotrophic microbes in the open ocean?

Answer: *Bacteria* that contain proteorhodopsin can obtain energy and therefore persist in the open ocean even when limited by carbon substrates. Other heterotrophic microbes have a more restricted metabolic capacity and therefore are at a disadvantage to proteorhodopsin-containing cells when carbon is the limiting nutrient.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 20.11

9) Explain why biofilms are a concern in medicine.

Answer: Organisms on the surface of a biofilm can be difficult to break apart due to the presence of an extracellular matrix, which means antimicrobials are often not as effective. Another issue is the organisms embedded within the biofilm are protected from antimicrobial exposure, so they are especially well protected and capable of surviving an antimicrobial treatment. This also means it is also very difficult for an immune system to rid of a biofilm.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.4

10) Describe four environmental conditions that would likely change the relative abundances of the members within a microbial community.

Answer: Changes to any of these six would likely perturb a microbial community: light abundance, osmotic conditions, oxygen content, temperature, pH, and water potential.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 20.1